

23CSE452 Business Analytics

Case Study

Decoding Smartphone Distraction: Analysing Notification and Usage Patterns

by

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1. Problem Statement and Objectives

1.1 Problem Statement

Smartphone distraction is often summarized using a single measure such as daily screen time. However, screen time does not capture how often a user checks the device, how many notifications are received, whether the user checks the phone without a new notification, or how frequently smartphone activity interrupts study or work. Two users with similar screen time can therefore have substantially different distraction patterns.

The objective is to analyse notification and usage behaviour, identify meaningful behavioural segments, and translate those segments into practical digital-wellbeing recommendations.

1.2 Objectives

- Analyse smartphone usage and notification behaviour using questionnaire-based data.
- Identify relationships among usage intensity, checking behaviour, interruptions, concentration and perceived productivity change.
- Use K-means clustering to discover and profile smartphone distraction personas and derive segment-level recommendations.
- Use a classification model such as Random Forest to predict frequent study/work interruptions and evaluate its performance using standard classification metrics.

1.3 Business Relevance

The analysis can support digital-wellbeing applications, educational institutions and workplace productivity initiatives by moving from generic advice such as “reduce screen time” towards behaviour-specific interventions. A user whose main issue is notification exposure may require a different intervention from a user whose main pattern is habitual checking or high screen time.

2. Data Collection and Dataset Description

2.1 Data Collection

The primary data source was the project questionnaire titled “Smartphone Usage & Distraction Behaviour Survey”. Initially, approximately 140 responses were collected through the questionnaire. These responses were then used as the basis for preparing the larger 10,000-record analytical dataset used in the final analysis.

<https://forms.gle/4gDcdHir3kRPoUb86>

2.2 Dataset Attributes

Category	Important variables	Role in analysis
Demographic context	Age, occupation	Descriptive analysis
Usage intensity	Daily screen time, phone checks	Core clustering variables
Notification behaviour	Daily notifications, work/study notification %, social/entertainment notification %	Core clustering and association variables
Checking behaviour	Checking without a new notification	Behavioural segmentation
Distraction	Study/work interruption frequency	Segmentation and association analysis
Outcomes	Concentration, perceived productivity decrease	Association analysis and cluster profiling
Digital wellbeing	Screen-time monitoring, app limits, Focus Mode, DND, notification management, effectiveness	Descriptive analysis and recommendations

3. Data Preparation and Exploratory Analysis

3.1 Data Preparation

- Removed fields not required for behavioural modelling, such as timestamp, age, occupation, email-fields where present.
- Cleaned text encoding artefacts and standardized column names.
- Converted range-based responses such as screen-time and notification ranges to representative numeric midpoint values.
- Converted ordinal response categories into consistent numerical levels for association analysis.
- Treated digital-wellbeing feature usage as categorical information and additionally derived a feature-count representation.
- Replaced non-finite numerical values and handled missing numerical values using the median before modelling.
- Standardized clustering variables using z-score scaling so that variables with different units would contribute comparably to K-means distance calculations.

3.2 Descriptive Analysis

Variable	Mean	Std.	Min	Median	Max
Screen time (hours/day)	4.75	2.26	1	5	9
Phone checks/day	47.24	25.49	10	50	110
Notifications/day	42.83	33.21	10	35	120

Variable	Mean	Std.	Min	Median	Max
Phantom checking	1.81	0.98	0	2	4
Work/study notification %	24.88	21.84	0	26	76
Social/entertainment notification %	26.63	22.67	0	26	76
Study/work interruptions	2.37	0.88	0	2	4

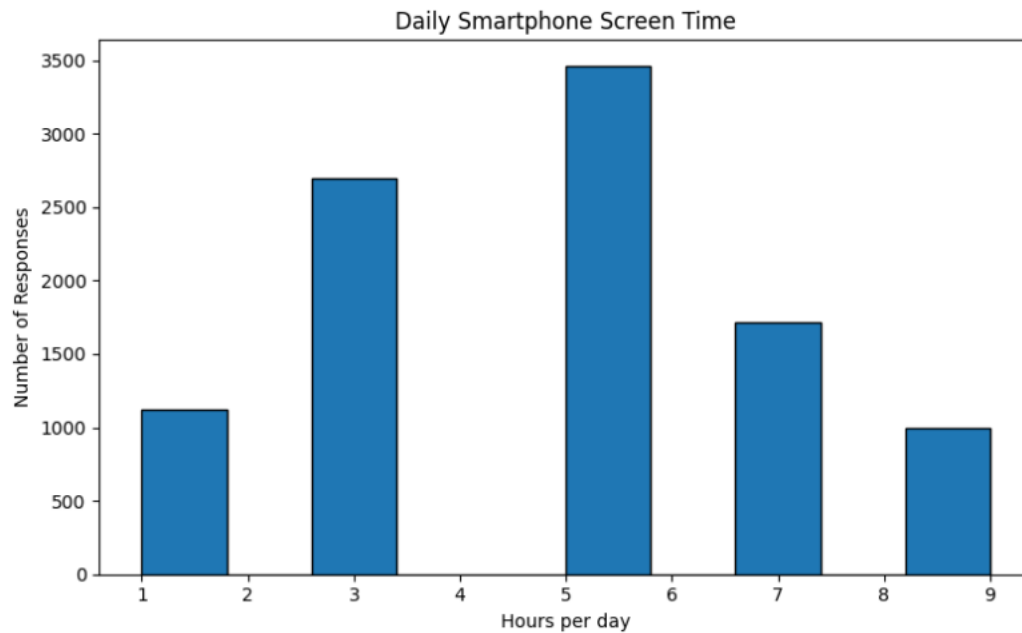


Figure 1. Distribution of approximate daily smartphone screen time.

3.3 Additional Exploratory Visualizations

The following plots were generated from the survey responses to get a simple view of the main usage and distraction patterns. They are included to make the distribution of responses easier to understand before applying the analytical methods.

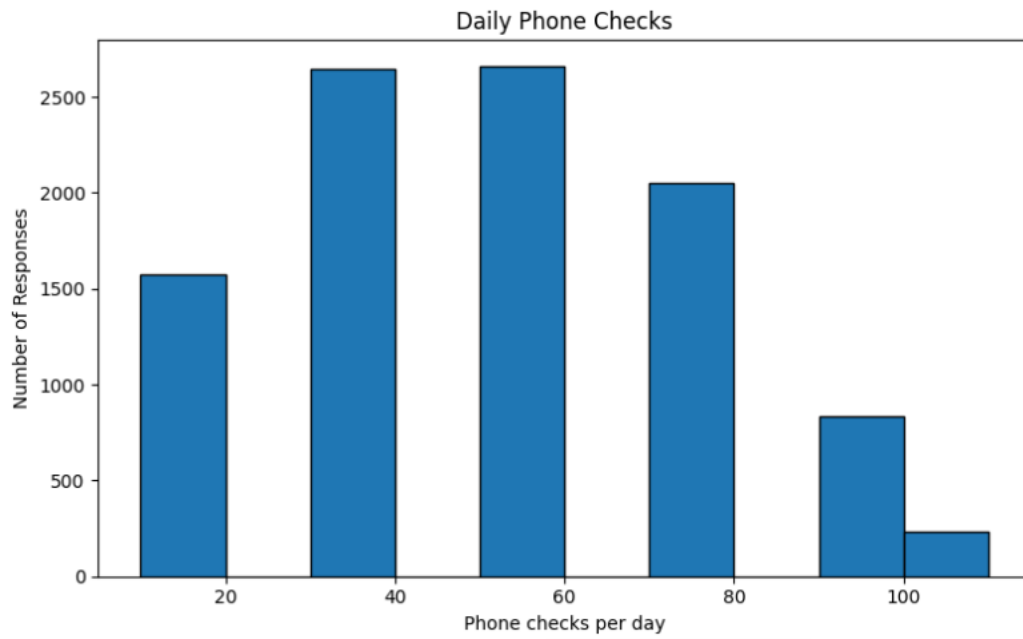
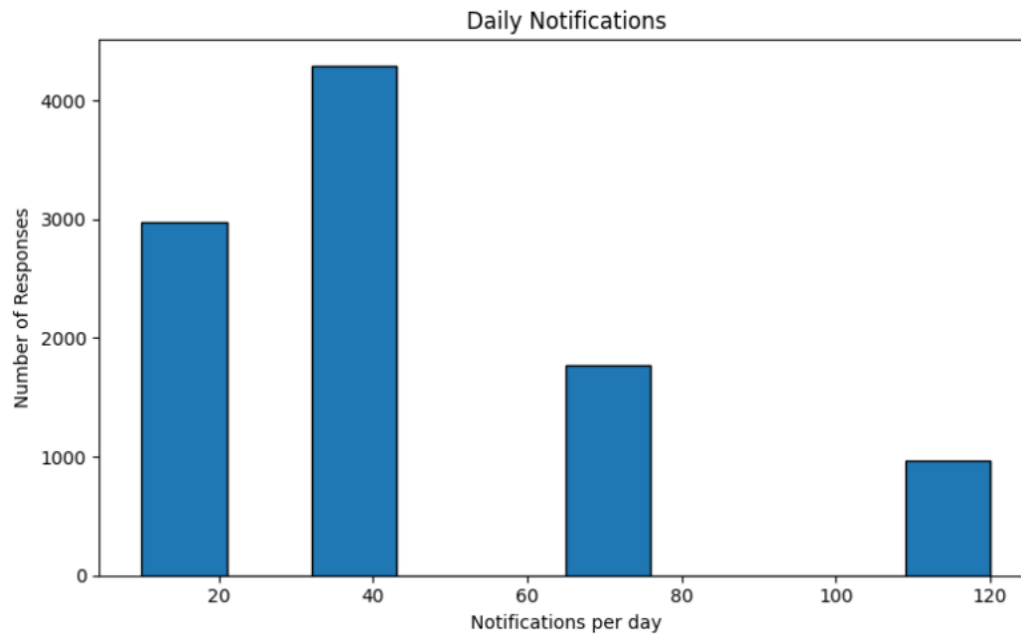


Figure 2. Daily smartphone usage and phone-checking frequency in the survey.



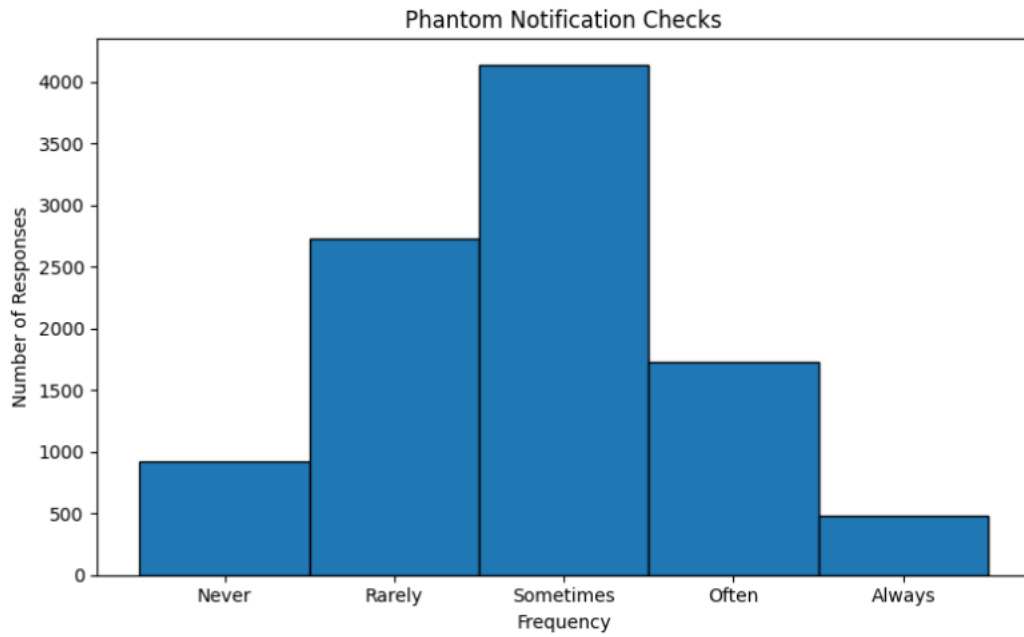
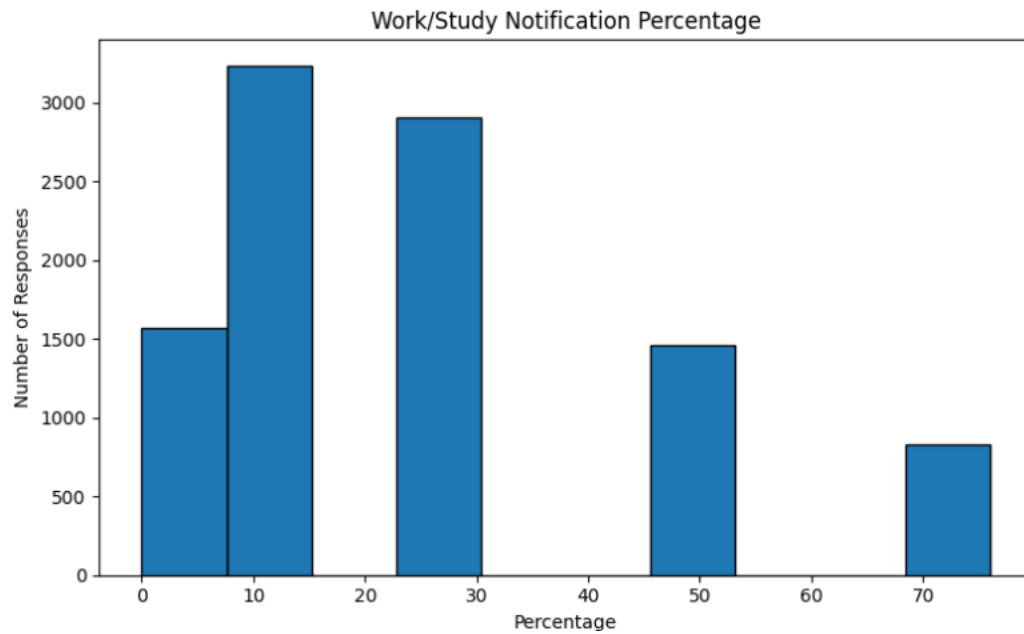


Figure 3. Daily notification levels and how often users check without a new notification.



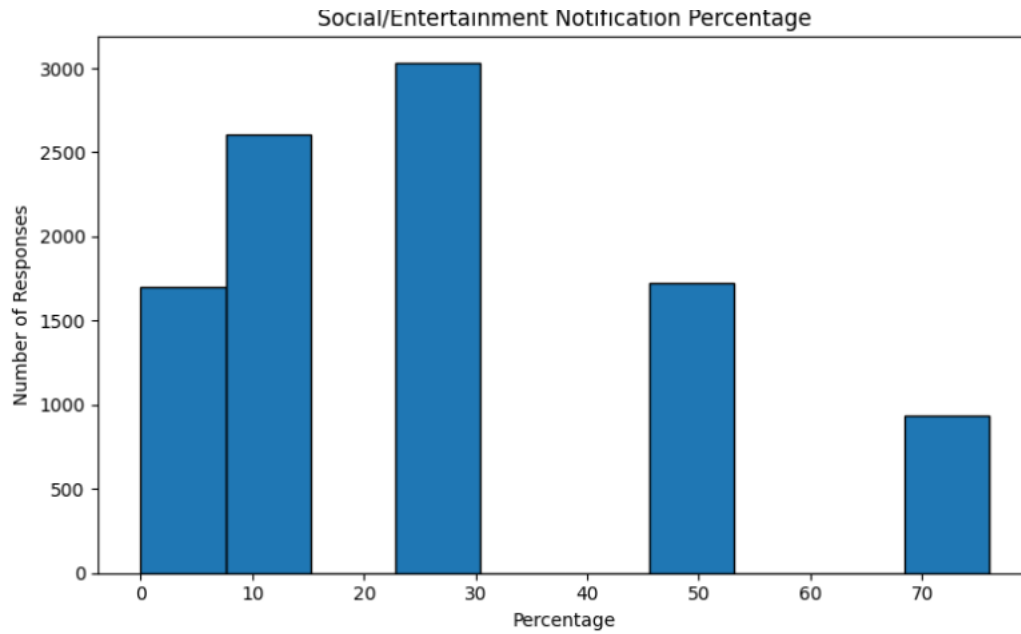


Figure 4. Distribution of work/study and social/entertainment notifications.

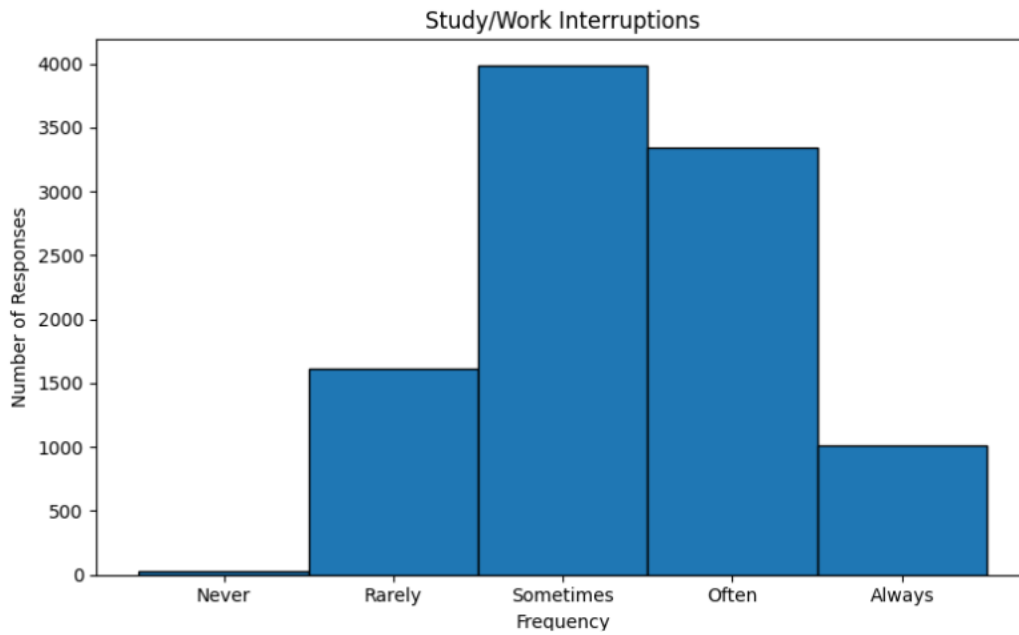


Figure 5. Frequency of study/work interruptions caused by phone notifications.

4. Analytics Method and Implementation

4.1 Association Analysis

Spearman rank correlation was used to examine monotonic relationships between behavioural variables.

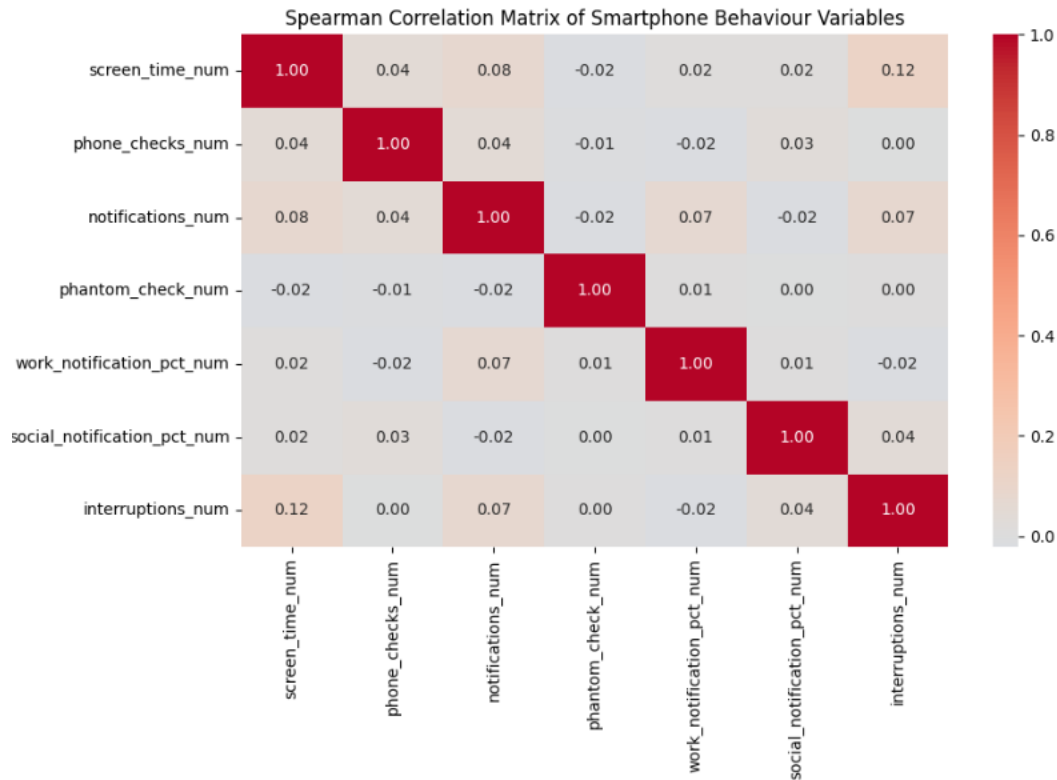


Figure 1. Spearman correlation matrix of the principal behavioural variables.

4.2 K-Means Clustering

K-means clustering was used to identify groups of users with similar smartphone usage and notification behaviour. The method was chosen because the main aim of the study is to discover behavioural segments rather than assign predefined labels to users. The clustering variables included screen time, phone checks, notifications, phantom checking, work/study notification percentage, social/entertainment notification percentage and study/work interruptions. These variables were standardized before applying K-means so that differences in scale would not disproportionately affect the clustering.

4.3 Choosing the Number of Clusters

Candidate values from $K = 2$ to $K = 6$ were evaluated using the silhouette score.

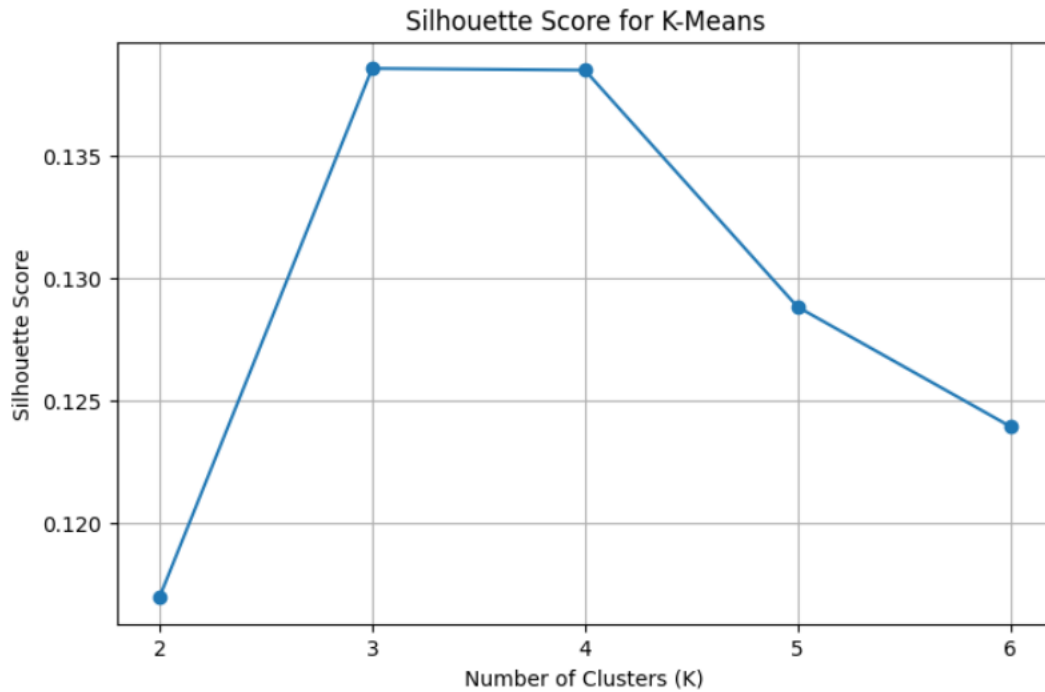


Figure 4. Silhouette scores for candidate K values.

K	Silhouette score
2	0.1170
3	0.1386
4	0.1385
5	0.1288
6	0.1240

$K = 3$ produced the highest silhouette score of 0.1386. However, $K = 2$ was retained for the final behavioural analysis because it provides two broad and easier-to-interpret distraction segments.

4.4 Classification and Evaluation

A classification step was included to identify users reporting frequent study/work interruptions. Users in the higher-interruption group were used as the target class, and the models were evaluated on a held-out test set.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.570	0.511	0.320	0.394
Random Forest	0.581	0.521	0.477	0.498

5. Comparison with State-of-the-Art Methods

Recent research approaches smartphone distraction from several perspectives, including configurational analysis, cognitive/EEG experiments, systematic review and meta-analysis, and behavioural intervention. These studies are complementary rather than directly comparable because their datasets, populations and evaluation designs differ.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with my work
Zhou & Deng (2024)	563 Chinese university students; interviews with 15 undergraduates	Semi-structured interviews + fuzzy-set qualitative comparative analysis	Configuration consistency / coverage	Identified four combinations of predictors associated with in-class smartphone distraction.	This study also seeks multi-factor behaviour profiles, but uses questionnaire data + K-means to create unsupervised segments. Our output is exploratory and not a causal/configurational model.
Electrophysiological effects of smartphone notifications (2024)	87 participants; randomized mindfulness/control conditions	EEG spectral analysis during a visual oddball cognitive-control task	Theta, alpha/beta power and theta/beta ratio	Notification exposure altered neural markers of attentional shifting; mindfulness buffered some cognitive-control effects.	This study measures self-reported notification and checking behaviour at population level rather than neural responses in a controlled experiment.
Paterna et al. (2024)	29 studies; 48,490 participants; 33 effect sizes	Systematic review + random-effects meta-analysis + meta-regression	Pooled correlation effect size (r), heterogeneity and moderator analyses	Found a small negative association between problematic smartphone use and academic achievement ($r = -0.110$).	This study focuses on behavioural segmentation and notification/checking patterns rather than estimating a pooled causal/effect size.
Brailovskaia et al. (2024)	278 employees in Germany; four intervention groups	Randomized experimental intervention with repeated measures	ANOVA, pairwise comparisons and effect sizes	Reducing non-work smartphone use was associated with improved work satisfaction, motivation and work-life balance.	This work is observational and survey-based. It identifies segments and recommendations but does not experimentally test intervention effectiveness.

6. Results, Business Insights and Recommendations

6.1 Association Results

Variable 1	Variable 2	Spearman ρ	p-value	Interpretation
notifications	phone checks	0.479	0.0000	Significant
phone checks	interruptions	0.108	0.2464	Not significant
phantom check	interruptions	-0.071	0.4424	Not significant
screen time	productivity decrease	-0.047	0.6141	Not significant
notifications	productivity decrease	-0.249	0.0066	Significant
interruptions	concentration	-0.315	0.0005	Significant

The strongest positive association in the supplied results is between notifications and phone checks. Notifications and perceived productivity decrease also show a statistically significant association under the current ordinal encoding. Interruptions and concentration show a statistically significant negative association. These results are associative and should not be interpreted as evidence that notifications or interruptions directly cause the reported outcomes.

6.2 Cluster Profiles

Cluster	Screen time	Checks	Notifications	Phantom check	Work notif. %	Social notif. %	Interruptions
0	3.83	44.78	34.51	1.82	24.98	23.86	1.79
1	5.83	50.11	52.50	1.81	24.76	29.85	3.04

6.3 Persona Interpretation

Segment	Observed behavioural characteristics
Cluster 0 – Lower Smartphone Distraction	Lower screen time, fewer notifications and lower study/work interruption levels, with comparatively less social/entertainment notification exposure.
Cluster 1 – Higher Smartphone Distraction	Higher screen time, more frequent phone checking, higher notification exposure and noticeably higher study/work interruptions.

6.4 Business Recommendations

Observed pattern	Recommended action	Business rationale
High notification exposure + frequent checking	Notification batching and priority filtering	Reduces unnecessary alert-driven device interactions while preserving critical alerts.
Higher screen time	Screen-time monitoring and app limits	Provides a measurable control mechanism for high-use segments.
Higher interruption levels	Scheduled focus periods and context-aware Do Not Disturb	Protects study/work blocks from avoidable interruptions.
Habitual/phantom checking	Intentional-check prompts and notification minimization	Targets checking behaviour that is not necessarily triggered by a new notification.
Work/study-heavy notification exposure	Separate essential work alerts from low-priority notifications	Maintains important communication while reducing non-essential attention switching.

6.5 Proposed Digital-Wellbeing Solution

A practical application based on this analysis could implement a lightweight behavioural profiling engine. The system would collect non-sensitive usage summaries, assign users to an exploratory behavioural segment, and provide segment-specific interventions. For example, a notification-heavy profile could receive batching and priority-filter suggestions, while a high-screen-time profile could receive app-limit and scheduled-break recommendations. The model should be presented as a decision-support mechanism rather than a diagnostic system.

7. Conclusion

This case study demonstrates how questionnaire data can be transformed into actionable business analytics for smartphone distraction. The analysis combines data preparation, exploratory analysis, association testing, classification and K-means clustering to examine smartphone usage as a multidimensional behavioural phenomenon.

The supplied results indicate a clear association between notification exposure and phone-checking frequency, while interruption frequency is associated with concentration. K-means produces two exploratory behavioural segments, although the silhouette scores indicate overlap between the groups. Consequently, the project emphasizes transparent interpretation and segment-specific recommendations rather than claiming that the clusters are definitive psychological personas.

The main business implication is that digital-wellbeing support can be more targeted when it considers notification pressure, checking behaviour, screen time and interruption patterns together. A larger verified dataset and objective device-usage data could be used in future work to improve the reliability of these segments.

References

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